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Factors influencing the Holocene history of *Fagus*

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ABSTRACT

The Holocene histories of three species of *Fagus* with related autecologies (*F. sylvatica*, *F. grandifolia* and *F. crenata*) are reviewed to compare factors that have affected the long-term dynamics of the species distributions. There are similar climatic controls exerted on all three species and their spreading histories were strongly influenced by the location of glacial refugia and the apparent establishment of outlying founder populations early in the Holocene. Under ideal growth conditions all three species showed similar population doubling rates, with estimated spreading rates of less than 100 m per year. Stand-scale disturbance is shown to have facilitated the spread of *F. sylvatica* at its northern distribution limits.

1. Introduction

Fagus is a widespread genus of ecological importance in eastern North America, Europe, China and Japan that has left a useful record of its distribution during the Holocene in the form of pollen grains and macrofossils preserved in sediments (Huntley and Birks, 1983; Peters, 1997; Tsukada, 1982a; Woods and Davis, 1989). This record can yield insight into how and why tree distributions changed through time and which other trees grew with *Fagus*. These topics are of potential importance for silviculture and conservation, particularly at a time of increasing concern about rapid climate change and its impact upon biota (Davis and Shaw, 2001; Hu et al., 2009). There have been numerous studies at the continental scale that have mapped *Fagus* distributions during the Holocene and interpreted the drivers of change (Bennett, 1988; Giesecke et al., 2007; Gliemeroth, 1995; Kito and Takimoto, 1999; Magri et al., 2006; Pott, 2000; Tsukada, 1982a; Woods and Davis, 1989). Huntley and Webb (1989) made an explicit intercontinental comparison of Holocene *Fagus* dynamics and concluded that the distributions of the closely related species of *F. sylvatica* and *F. grandifolia* were both responding to long-term climate change, particularly late-Holocene increases in winter temperatures (Huntley et al., 1989; Huntley and Webb, 1989). However there are several other factors that also potentially impact the long-term dynamics of species distributions, such as location and properties of glacial refugia (Bennett and Provan, 2008), rates of spreading and population increase (Bennett, 1986; Clark et al., 1998), competitive relationships with other tree species, disturbance regime and human influence. There are likely to be interactions between these different factors. In this paper we consider the Holocene record of three species of *Fagus* (*F. sylvatica*, *F. grandifolia* and *F. crenata*) growing in Europe, North America and Japan. We analyse aspects of their Holocene records to see how the past can contribute to understanding the relationships between these various properties and processes that are of importance for *Fagus*.

2. Autecology and modern distributions

F. sylvatica has a typical lifespan of about 300 years and can reach a height of more than 40 m (Ellenberg, 1996). It has a shallow root system that makes it susceptible to drought (Peterken and Mountford, 1996) and wind-throw and its thin bark provides little protection from forest fires. *F. sylvatica* starts flowering at an age of 40–50 years in the open and 60–80 years in dense stands (Firbas, 1949). Reproduction is primarily sexual and high seed production (masting) occurs irregularly, often triggered by drought in the previous year (Piovesan and Adams, 2001). Seed dispersal over longer distances relies on transport by animals such as squirrels and jays as well as by streams. Vegetative reproduction through layering only occurs in montane regions. Seed germination occurs in spring. Adequate moisture availability is very important for germination and the young seedlings are easily killed by drought or late frost (Watt, 1923). *F. sylvatica* of all ages is particularly

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susceptible to late frost (Grime et al., 1988; Lang, 1994). To protect the sprouting leaves, the tree has developed a high level of dormancy, and requires either a long period with temperatures > 5 °C, or a long chilling period to initiate budburst (Murray et al., 1989). Many European forests currently experience rather intense browsing pressure from ungulates, but mature *F. sylvatica* is relatively unpalatable which is a contributory factor to its local dominance (van Hees et al., 1996). *F. sylvatica* avoids waterlogged and dry sites, but is a vigorous, shade-tolerant and shading competitor on a wide range of soil types, which allows the species to achieve dominance or co-dominance in many habitats, making it one of the most important forest trees in Europe often forming large, monospecific stands (Ellenberg, 1996).

The distribution of *F. sylvatica* (Fig. 1) extends from southern England and southern Sweden, where it is a lowland tree, to the Apennines of southern Italy, where it occupies an altitudinal belt between 1100 m and 1900 m (Jalas and Suominen, 1972–1999). *F. sylvatica* requires moist summers and mild winters and is thus absent from continental areas in Eastern Europe.

F. grandifolia shares many of the ecological characteristics of *F. sylvatica* outlined above. Important differences are its occurrence in more continental climates (Fang and Lechowicz, 2006) and the role of blue jays (and formerly passenger pigeons) that may disperse seeds for several kilometres (Darley-Hill and Carter, 1981). Unlike *F. sylvatica*, *F. grandifolia* is capable of vegetative reproduction from root suckers or stump sprouts (Clark, 1991). Individuals of *F. grandifolia* can live longer than many species with which they co-occur (Poulson and Platt, 1996). Community-level analyses of temperate forests have suggested that *F. grandifolia* usually reaches the canopy in small gaps. Relative abundance of this species has been proposed to increase during periods with a low rate of canopy disturbance, typically caused by hurricane damage, and decline when such disturbance is more frequent (Peters and Platt, 1996). However demographic analyses have shown that *F. grandifolia* has a strategy for surviving canopy disruption following hurricanes through survival of large understorey individuals (Batista et al., 1998). Fire has been an important ecological feature of North American deciduous forests and *F. grandifolia* is very vulnerable to fire damage. Postfire colonisation is through root suckering (Swan, 1970). *F. grandifolia* can become dominant or co-dominant with a range of deciduous genera (*Acer*, *Betula*, *Carya*, *Prunus*, *Quercus*, *Tilia*) and *Pinus strobus*. It rarely forms monospecific stands, although isolated pure stands have developed in southern New England following hurricanes when establishment of potential competitors has been restricted by changes in land management (Busby et al., 2009a,b).

F. grandifolia is found at low elevations in the north and relatively high elevations in the southern parts of its range (Fig. 2) (Braun, 1950). Local soil and climatic factors probably determine whether *Fagus* grows at the higher elevations. In the Adirondack Mountains, low temperatures and wind keep *F. grandifolia* below 975 m in contrast to the Appalachian Mountains where on the warmer slopes it grows at elevations up to 1830 m. At altitudes in the middle of its range, *F. grandifolia* is more abundant on the cooler, moister, northern slopes than on the southern slopes (Tubbs and Huston, 1990).

F. crenata is a dominant tree species in cool temperate forests of Japan and as with the other *Fagus* species has a strong masting habit. *F. crenata* can comprise the majority of the biomass in a Japanese cool temperate forest (Nakashizuka, 1984). Its masting habit has considerable effects on forest regeneration processes (Kon et al., 2005) and the population dynamics of animals that feed on *F. crenata* seeds (Hoshizaki and Miguchi, 2005). Different types of *F. crenata* forest occur along a snow accumulation gradient. Sasaki (1970) and Hukushima et al. (1995) divided these forests into two types: Japan Sea (JS) and Pacific Ocean (PO). JS-type beech

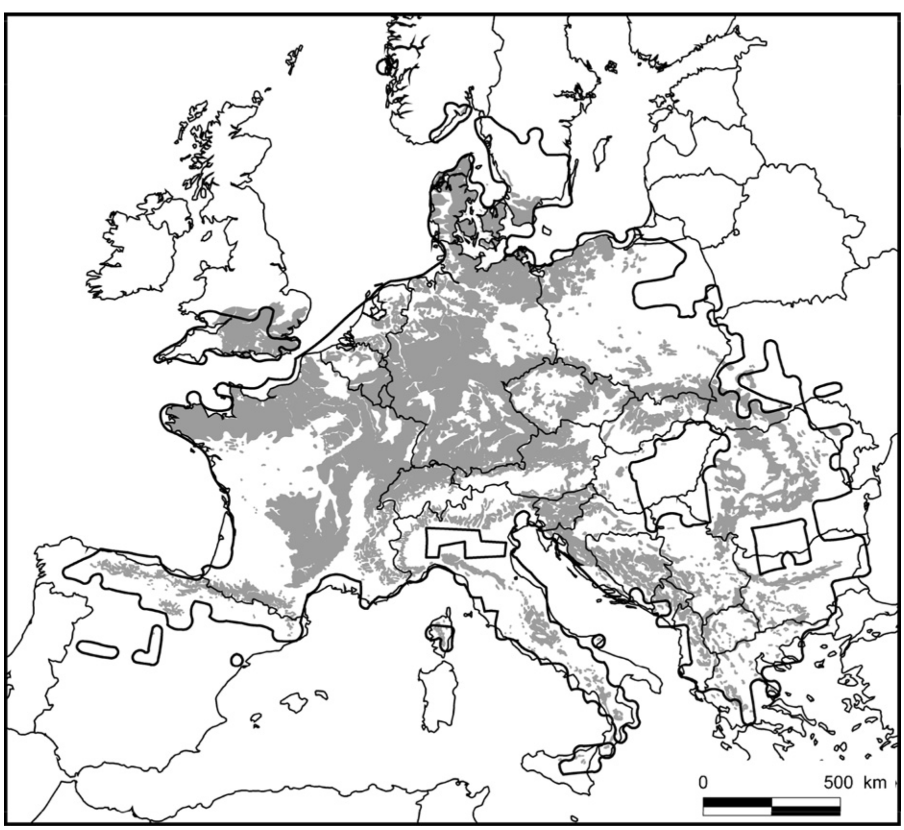

Fig. 1. Modern natural distribution of *Fagus sylvatica* according to Atlas Florae Europaeae (Jalas and Suominen, 1972–1999), bold black line; and areas where *Fagus sylvatica* would be the dominant tree under natural conditions (Bohn et al., 2000), shaded (after Giesecke et al., 2007).

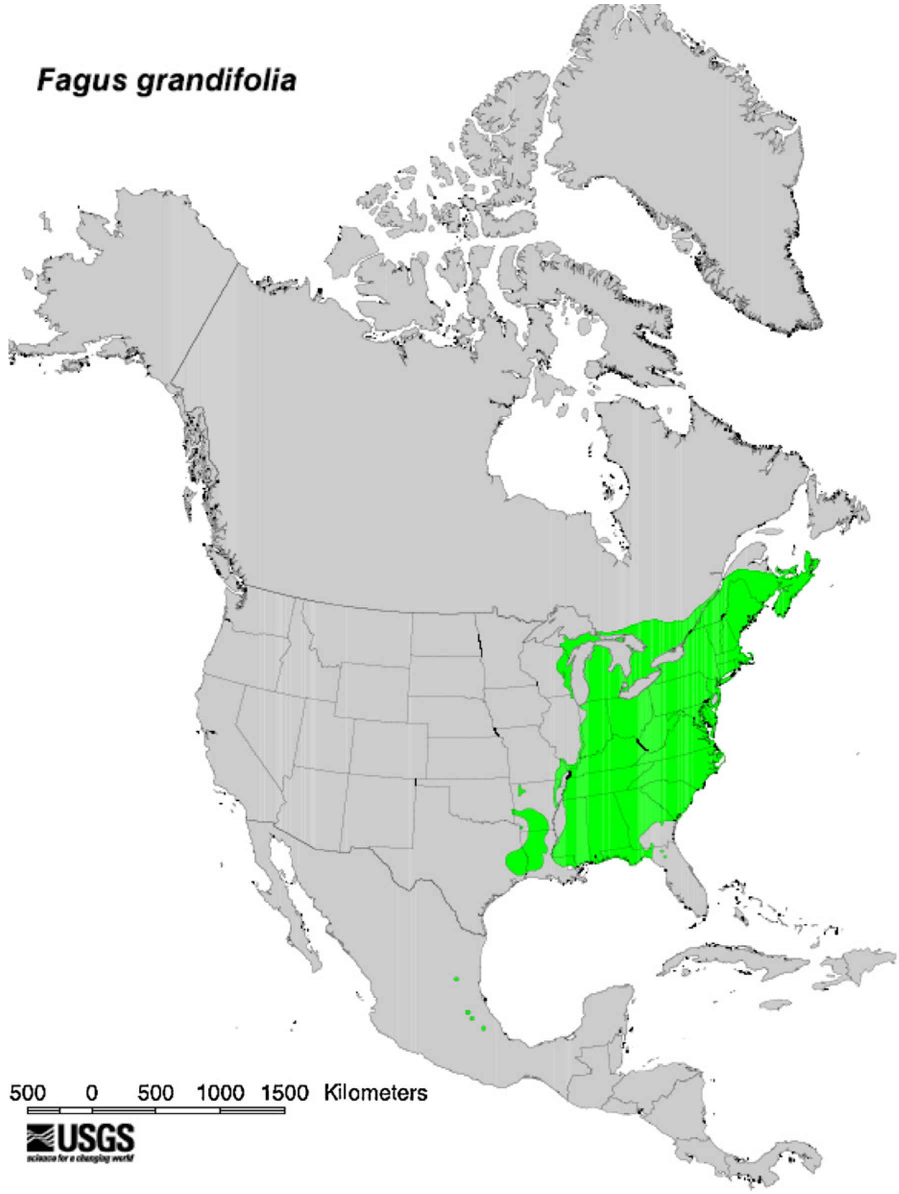

Fig. 2. Modern distribution of *Fagus grandifolia* in North America (United States Geological Survey).

forests occur in the northern and western part of the Japanese archipelago, near the Japan Sea, whereas PO types occur in the south, facing the Pacific Ocean (Fig. 3). JS-type beech forests are distributed in snow-rich areas and are associated with some characteristic plants, such as *Aucuba japonica*, *Hamamelis japonica*, *Daphniphyllum macropodum*, *Skimmia japonica* and *Sasa kurilensis* (Shimano, 2002). These forest floor shrubs, especially evergreen shrubs, can occur in the cool temperate zone because the snow cover protects them from cold air temperatures. This type of forest is almost pure *F. crenata* forest because of the elimination of other trees by snow (Shimano, 2002). PO-type beech forests occur in areas with less snow and have many co-dominant trees (Shimano and Okitsu, 1994). Regeneration of natural *F. crenata* forests has been explained by the gap dynamics theory in Japan (Yamamoto, 1989). It is assumed that *F. crenata* forests have various patches of different developmental stages, resulting in a mosaic pattern; the seedling bank regenerates with canopy gap formation; the growing saplings become canopy trees that undergo competition for space with neighbouring trees; and the climax *F. crenata* forest in a patch mosaic is stable as a regenerating complex. In addition, the timing of beechnut masting and its relationship to the death of dwarf bamboo (*S. kurilensis*) after flowering has been discussed (Shimano, 2002). Heavy snow accumulation flattens dwarf bamboo culms, which improves forest floor light conditions for beech seedlings.

3. Climatic controls

The present distributions of all three species are believed to be under a broad climatic control (Fang and Lechowicz, 2006; Huntley et al., 1989; Pearman et al., 2008). The approaches taken to demonstrate this control relate current distribution limits to modern climatic variables. These approaches make the important assumption that current distributions are controlled by and in equilibrium with modern climate. The generally good fit between the spatial distribution of potentially limiting climate parameters and species distribution limits is also seen in response surfaces (Bartlein et al., 1986) and has lent support to this assumption. Huntley et al. (1989) used modern pollen data as a proxy for *Fagus* distributions in North America and Europe. They demonstrated that response surfaces, depicting *Fagus* pollen percentages in

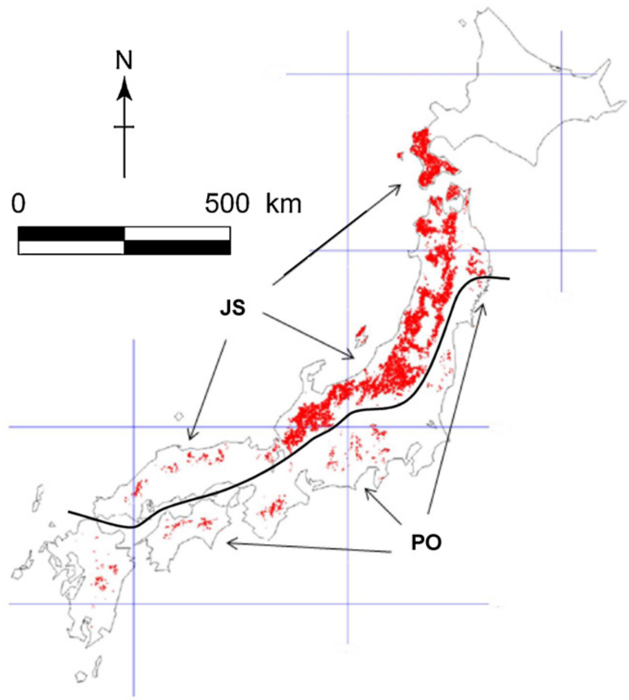

Fig. 3. Modern distribution of *Fagus crenata* in Japan. JS, Japan Sea forest type; PO, Pacific Ocean forest type (after Report from Forestry and Forest Products Research Institute (FFPRI), Tsukuba, Japan).

surface samples plotted against mean January and July temperatures of the sampling sites, were similar on both continents (Fig. 4). *Fagus* pollen was most abundant when mean January temperatures lay between −1 and −4 °C and mean July temperature was 18 °C (Huntley and Webb, 1989). These observations have been largely supported by measurements made at the distribution limits of 11 *Fagus* species around the world (Fang and Lechowicz, 2006). Their study confirmed that temperature was the most important factor controlling *Fagus* distribution worldwide, with moisture effects subsidiary. They identified slightly different climatic tolerances for each species, for example *F. grandifolia* occurred in warmer conditions than other species, *F. sylvatica* was more drought tolerant than *F. grandifolia* and *F. crenata*, with the latter growing in the wettest conditions. These authors also identified small regions of apparently favourable climate not yet occupied by *Fagus* species, for example in northern Britain and Yakushima Island, Japan.

This rather detailed knowledge of the climatic tolerances of *Fagus* has been incorporated into several types of model, often with the aim of forecasting the effects of future climate on species distribution (Koca et al., 2006; Pearman et al., 2008). Sykes et al. (1996) combined data on winter cold tolerance, length of the growing season and winter chilling requirements with drought tolerance to develop bioclimatic variables based on physiological constraints. They built the equilibrium model STASH (STAtic SHell) to determine the presence of a given species within a grid cell. Their model gave a good fit for *F. sylvatica* to the observed modern distribution (Sykes et al., 1996).

The corresponding distribution of *F. sylvatica* in Europe was computed for 6000 years ago (all dates are calibrated ^{14}C years), using the same model (Giesecke et al., 2007) (Fig. 5). This model run was driven by probable climate data for 6000 years ago derived from atmospheric general circulation model (AGCM) outputs. The modelled distribution of *F. sylvatica* based on climate data from three different AGCMs yielded distribution patterns similar to the present occurrence of the tree. On the other hand palaeoecological data showed clearly that the regions in which *F. sylvatica* trees were common at the time lay to the south, including the forelands of the Alps and parts of the Balkan peninsula. Climate model results with 2–6 °C warmer average winter temperatures gave rise to the simulation of *F. sylvatica* stands further into northeast Europe than at present. Scenarios with higher summer temperatures imposed some drought stress, which restricted the simulated range in southern Europe and created poor growing conditions in western France.

Possible reasons for this data–model mismatch included uncertainties of the palaeoclimatic estimates, in particular possible changes in seasonality (Giesecke et al., 2008). Also the focus on average climate change rather than on changes in the frequency of events such as late frost may have biased the results. The hypothesis explaining the late spread of *F. sylvatica* with migrational lag was also demonstrated not to be fully consistent with the data. For example, *F. sylvatica* survived the last glaciation in the Balkans and other locations around the Mediterranean (Magri et al., 2006) but did not become a dominant species in Bulgaria and many other regions until late in the Holocene (Tonkov, 2003). The expansion of *F. sylvatica* populations often coincides with disturbance events, which may be climatic in nature such as the event 8200 years ago (Tinner and Lotter, 2006). Shuman et al. (2006) reported a century-scale cool period in the northeastern USA during this trans-Atlantic climatic event, after which lake levels rose and plant communities reorganized in favour of moisture-loving taxa that included *F. grandifolia*. Early Holocene climate in the northeastern USA appears to have been too

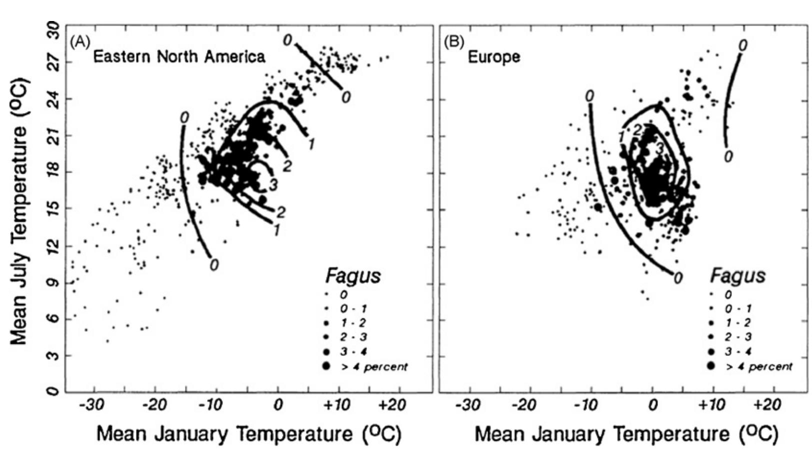

Fig. 4. Response surfaces: pollen abundance versus January and July mean temperatures: (A) North America and (B) Europe (Huntley et al., 1989).

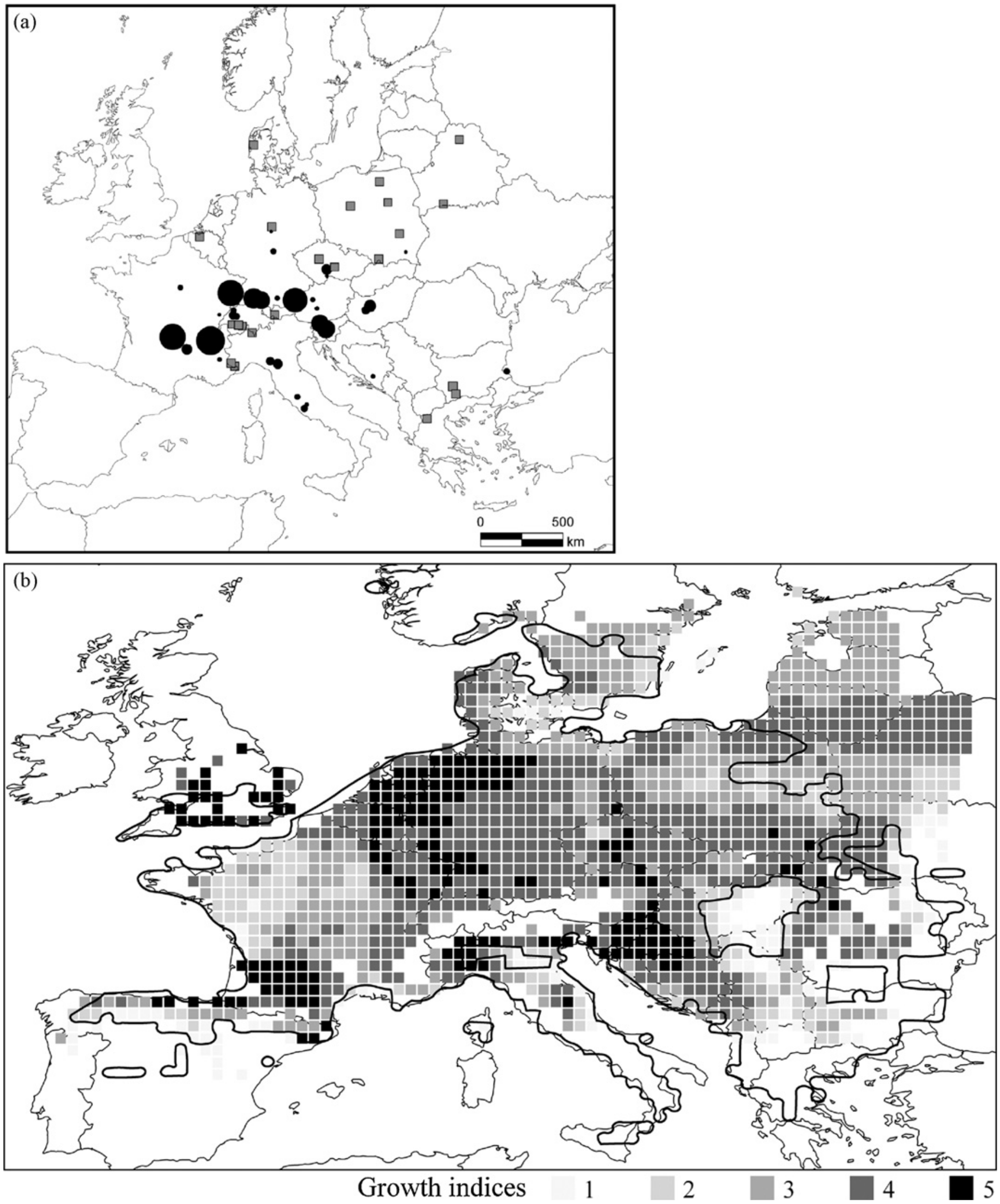

Fig. 5. (a) Percentage of *Fagus sylvatica* pollen in selected pollen diagrams across Europe at 5700 cal. BP. The size of filled circles is equivalent to the amount of pollen; grey squares indicate sites with *Fagus* pollen < 0.5% (after Brewer, 2002). (b) Simulation of the *Fagus sylvatica* distribution based on the ECHAM3 AGCM output of climate parameters 6000 years ago. The modern distribution of *F. sylvatica* according to Atlas Florae Europaeae (Jalas and Suominen, 1972–1999) is shown for comparison (bold black line) (after Giesecke et al., 2007).

arid for *F. grandifolia* and was associated with a wildfire regime that was unsuitable for the species (Faison et al., 2006). However climatic factors alone cannot yet be demonstrated to account fully for the Holocene history of *Fagus*. A combination of climate together with other processes such as spreading biology, population dynamics and chance disturbance deserve to be investigated. In the next section we explore possible relationships between climate and population dynamics.

4. Climate and population growth

Distribution limits of species are difficult to reconstruct using palaeoecological data owing to problems of pollen representation and dispersal (Davis et al., 1991) and poor sensitivity of standard investigation sites to small populations (Bennett, 1986). Pollen data are better suited to reconstruct gradients of dominance, and they also record rates of population increase where pollen accumulation rate data are sufficiently detailed and well dated (Watts, 1973). Estimates of *Fagus* population doubling times, which are the time taken for pollen values to double in size, can be calculated from the first half of the Holocene for a small number of sites in Japan (Tsukada, 1982a), Canada (Bennett, 1988) and Europe (Giesecke et al., 2007) (Table 1). These estimates not only indicate rates of population increase, but also help distinguish genuine species spread from a spatial gradient of decreasing doubling times that can be misinterpreted as a migrating front (Bennett, 1988; Kito and Takimoto, 1999). The Japanese doubling times fall into two groups. Four sites have doubling times of less than 250 years and one site of over 400 years. The fastest doubling time (Nojiri) is from the late glacial period when there was sparse forest cover, suggesting a combination of optimal climatic conditions with minimal competitive pressure (Tsukada, 1982a). The other sites

Table 1
Doubling times at various ages for *Fagus* populations.

Site	Doubling time (years)	Age (years BP)	Reference
Nojiri, Japan	126	12,000	Tsukada (1982a,b)
Soppensee, Switzerland	200	8000	Giesecke et al. (2007)
Meerfelder 2, Germany	200	4500	Giesecke et al. (2007)
Tashiro, Japan	210	7000	Tsukada (1982a,b)
Yakumo, Japan	230	3400	Kito and Takimoto (1999)
Mi-ike, Japan	231	10,000	Tsukada (1982a,b)
Nutt 2, Canada	242	5000	Bennett (1988)
Hams, Canada	256	8500	Bennett (1988)
Meerfelder 1, Germany	430	6500	Giesecke et al. (2007)
Ohyachi, Japan	433	9000	Tsukada (1982a,b)
Nutt 1, Canada	444	7500	Bennett (1988)
Bastien, Canada	444	8000	Bennett (1988)

with rapid doubling times are where *Fagus* grew in almost pure stands with little competition and apparently optimal climatic conditions.

Bennett (1988) questioned the migrational lag concept based on his estimates of population doubling rates in Canada. An apparent northward spread of the species could be attributed to different rates of population increase that were under climatic or competitive control. In Europe, doubling rates of 200 years are recorded at both Meerfelder Maar and Soppensee suggesting that under ideal climatic conditions, a common factor is restricting population growth (Table 1). Indeed the doubling rates on all three continents that are faster than 250 years suggest that there are biological limits to population growth for *Fagus* under near ideal conditions. These rates are however significantly slower than those recorded for *Picea abies* in Scandinavia (Giesecke, 2005). Rates of population expansion and geographical spread are likely to be linked, although there is growing evidence in the palaeoecological record for potentially long delays between initial colonisation and eventual population expansion (Gardner, 2002). In the next section we review the concept of spreading rate for *Fagus* during the Holocene.

5. Spreading rates and cryptic refugia

Migration rates or spreading rates of species have been a preoccupation of palaeoecologists and biologists for many years, not least because of concerns about range adjustments needed to track climatic changes forecast for the near future (Davis and Zabinski, 1992). Several palaeoecologists have calculated spreading rates for *Fagus* based on estimated arrival dates for the tree at multiple sites, but doubt has been cast on the validity of these estimates owing to the difficulty of interpreting first appearance of *Fagus* in an area using pollen data alone (Bennett, 1986; Davis et al., 1991; McLachlan et al., 2005). Pollen-based estimates of tree spreading rates have also been far more rapid (100s to 1000s of metres per year) than those observed in modern forests (10s of metres per year) or theoretically predicted, unless high rates of long-distance dispersal are invoked (Clark et al., 2001). Estimation of spreading rate is crucially influenced by the location of the 'starting' points of the spreading process and hence the distribution of the species during the last glacial maximum (LGM). Earlier work implied that these LGM distributions lay at or near the southern margins of the current ranges of *Fagus*, but recent palaeoecological and genetic studies in Europe have proposed some more northerly occurrences (so-called 'cryptic refugia') or early Holocene locations (Bhagwat and Willis, 2008; Petit et al., 2003; Stewart and Lister, 2002; Willis et al., 2000). Gardner (2002) presents a *Fagus* record from a peat deposit 200 m × 100 m in northeast Hungary that indicates a low but consistent presence of

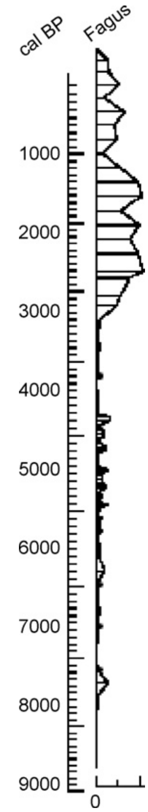

Fig. 6. *Fagus sylvatica* pollen percentage record from Sirok Nyírjes Tó, northeast Hungary (47°56′N, 20°11′E) (after Gardner, 2002).

Fagus for almost 9000 years (Fig. 6). Overballe-Petersen et al. (in prep.) have also recorded *Fagus* pollen and macrofossils covering the last 9000 years with a major population expansion only 1000 years ago from a Danish small forest hollow. Such sites record small populations with greater sensitivity than larger lake basins (Bradshaw, 2007).

Bialozyt et al. (in prep.) have modelled the spread of *F. sylvatica* into southern Scandinavia during the last few thousand years and obtained no satisfactory agreement between a diffusion model and observations of colonisation derived from palaeoecological records in small forest hollows. Given the real possibility of scattered, old founder populations of *Fagus*, the most reliable estimates of spreading rates might come from observed, active regions of spreading for the tree in forested landscapes such as southern Sweden (Bradshaw and Lindbladh, 2005). The spreading rate calculated from small hollow sites at the northeastern range limits of *F. sylvatica* is about 30 m per year.

The Japanese and Canadian data reviewed in the previous section all indicated an early, widespread dispersal and establishment of *Fagus* populations, followed by differential expansion rates related to initial population size and latitude (Bennett, 1988; Tsukada, 1982a). It seems reasonable to propose that long-distance dispersal, establishment and initial population expansion were likely in open, non-forested landscapes, with subsequent population growth dependent upon local climate and competitors. In Japan, the late-glacial rapid dispersal of *Fagus* is well documented on Honshu (Tsukada, 1982b) and may be attributed to spreading from cryptic refugia. On the other hand, the later and slower colonisation and spread on the northern island of Hokkaido was a simpler, standard migration process (Kito and Takimoto, 1999). The spread of *F. crenata* on Hokkaido during the last 6000 years has been estimated at 20 m per year (Fig. 7). This very slow rate of spread is also influenced by both competition from existing forest

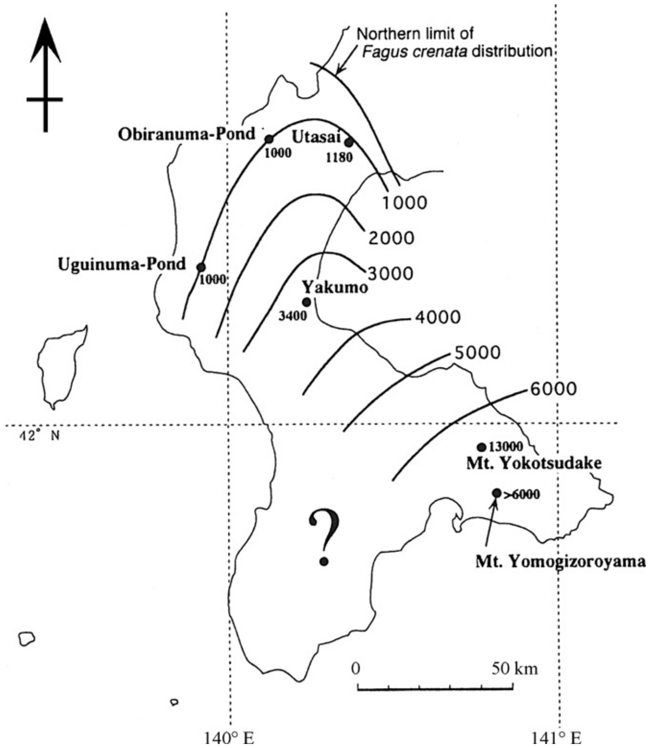

Fig. 7. Spreading map of *Fagus crenata* during the Holocene in southwestern Hokkaido, Japan. Isochrones are in years before present (after Kito and Takimoto, 1999).

trees and the inhospitable seed bed created by the dwarf bamboo *Sasa*, which only periodically permits *Fagus* regeneration following large-scale die-back of *Sasa* after flowering.

The concept of survival of *Fagus* populations close to ice-sheets has received strong support from the modern genetic structure of *F. grandifolia* populations in North America, where a significant proportion of non-adaptive diversity is found in northern parts of the modern range (McLachlan et al., 2005). These authors consequently estimate *Fagus* spreading rate to be 80–90 m per year, which is far lower than previous estimates based on pollen data alone. Clearly the estimation of past spreading rates is fraught with difficulty, but a consensus is beginning to emerge based upon comparison of dispersal biology observations with genetic and palaeoecological data. Potential rates appear to be less than 100 m per year, which are significantly lower estimates than those quoted from older literature by McLachlan et al. (2005). The continental scale spreading process deserves further study as earlier interpretations may not account for the importance of dispersed, early founding populations.

6. The role of disturbance and human impact in the spread of *Fagus*

Iversen (1973) argued that *F. sylvatica* needed natural or anthropogenic disturbance to spread because most of Europe was already covered by dense forest, within which it was difficult for invading trees to establish (Green, 1987). Subsequent work has indicated the type of anthropogenic disturbance that favoured the spread of *Fagus*, at least in northern Europe (Bolte et al., 2007; Küster, 1997; Odgaard, 1994). Periods of *Fagus* expansion are particularly associated with abandonment of temporarily cultivated regions (Andersen et al., 1983). *Fagus* populations tended to decrease as shifting cultivation, often associated with local burning, was replaced by more permanent field systems (Küster, 1997). A weak association between *Fagus* population expansion and Neolithic farming activities has been described from central Europe (Tinner and Lotter, 2006), but most palaeoecological studies from elsewhere in the *Fagus* distribution lack the resolution to assess the relationship between population dynamics and anthropogenic disturbance.

Bradshaw and Lindbladh (2005) examined the relationship between different aspects of the disturbance regime (fire and anthropogenic activity) and population size of *F. sylvatica* in southern Scandinavia, by comparing stand-scale pollen records with charcoal concentrations and pollen indicators of anthropogenic impact. Analysis of the relationship between pollen percentage values and charcoal data showed that charcoal values were generally higher immediately prior to local *Fagus* establishment, although after establishment, *Fagus* was disadvantaged by local fire. The stand-scale data also showed a significant association between presence of *Fagus* and pollen from cultivated cereals and other anthropogenic indicators (Bradshaw and Lindbladh, 2005). In a survey of 890 samples from stand-scale sites, they also found high percentages (>5%) of *F. sylvatica* pollen in 40 samples where the other pollen indicators of anthropogenic activity exceeded 10% of total pollen. *P. abies* was spreading into the region at the same time as *Fagus*, yet its pollen exceeded 2.5% only once in the same survey. These data strongly suggest that sustained *F. sylvatica* populations occur more commonly on sites with high anthropogenic influence than is the case for *P. abies*. This association would contribute to a greater patchiness in the temporal and spatial distribution of *F. sylvatica* and slow the spread of the species in comparison with *P. abies*: a pattern that is observed in the data from small forest hollows.

The spread of *F. sylvatica* in southern Scandinavia and northern Germany (Küster, 1997) has therefore been directly linked to specific anthropogenic activities such as cultivation and disturbance by fire prior to stand establishment. Thus earlier forms of land-use in this region, involving fire and cultivation of marginal land have favoured the spread of *F. sylvatica*. However comparable observations have not been made throughout its range, because high-resolution stand-scale sites have only been studied in northern Europe. Human activity has created openings in parts of European forests throughout the Holocene and its impact on vegetation has generally increased through time (Behre, 1988). Thus, if human activity was a key factor, thresholds in the scale of openings or a change in forest usage have to be considered. The general spread of Neolithic farming across Europe (Roberts, 1998) has the same direction as the spread of *F. sylvatica* as depicted by Gliemeroth (1995), and the greater magnitude of forest opening since 6000 years ago would agree with the time of spread and population expansion in northern Europe. However, there are a number of sites and regions where this argument does not hold (Gardner and Willis, 1999). Tinner and Lotter (2006) discussed pollen diagrams from the northern forelands of the Alps where the expansion of *F. sylvatica* occurred prior to significant human activity, although they suggested that the climatic 'event' 8200 years ago may have acted as a decisive disturbance. In contrast, the expansion of *F. sylvatica* in the mountains of Bulgaria as well as in northern Hungary occurred only after 5000 years ago (Gardner, 2002; Stefanova and Ammann, 2003; Tonkov, 2003) even though these regions have a long history of human impact, extending back almost 9000 years (Galaty, 2005). Human impact may have been an important local factor in population expansion and it may have helped the tree develop from presence to dominance in marginal regions, but it cannot fully explain the pattern of *F. sylvatica* spread and population expansion on the European scale. Human impact has not been considered to have been a significant issue in the spread of either *F. grandifolia* or *F. crenata*.

7. Concluding remarks

The recent combination of genetic data with more detailed palaeoecological records have necessitated a re-evaluation of earlier concepts of Holocene spread of *Fagus* from distant southern refugia on three continents. Rather than a continuous northwards spread constrained by climatic or biological factors, it appears that several outlying founder populations may have established early in the Holocene and acted as nuclei for further spread, when local climate become particularly favourable for *Fagus*. Where spreading rates can be calculated, they are of the order of 10s of metres per year, which may be too slow to track rapidly changing future climates. Disturbance has been associated with the past spread of at least *F. sylvatica* at the northern margins of its distribution.

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